

JRAHS Year 12 Extension 2 Term 2 Assessment 2011

QUESTION 1 (15 Marks)

Marks

(a) A rectangular hyperbola has equation $xy = a^2$

(i) Find the equation of the tangent at the point $P\left(at, \frac{a}{t}\right)$ on the hyperbola.

2

(ii) The tangent at P meets the x and y axes at L and M respectively. O is the origin and POQ is a diameter. The line MQ meets the x -axis at T.

4

Prove that the area of $\Delta QOT = \frac{a^2}{3}$ units²

(b) (i) If a particle moving with velocity v experiences air resistance equal to kv^2 per unit mass (k being constant), prove that, in falling from rest in a vertical line through a distance s , it will acquire a velocity of v given

$$\text{by } v = V\sqrt{1 - e^{-2ks}}$$

3

where $V = \sqrt{\frac{g}{k}}$ (the terminal velocity) and g is the acceleration due to gravity (assumed constant).

(ii) With the same air resistance as in (i) acting on a particle, show that if projected upwards with velocity U , it will reach a height x , given by

3

$$x = \frac{1}{2k} \ln \left(\frac{g + kU^2}{g} \right)$$

(iii) Hence prove that under the same conditions, a particle projected upwards with velocity U will return to the point of projection with velocity W given by

3

$$W^{-2} = U^{-2} + V^{-2}$$

QUESTION 2 (START A NEW PAGE) (15 Marks)**Marks**

(a) A particle of unit mass is projected vertically upwards against a constant gravitational force g and a resistance $\frac{v}{c}$, where v is the velocity of the particle and c is a constant and s is the distance travelled in time t ; at $t = 0$, $s = 0$, and $v = c(h - g)$ where h is a constant.

(i) Write down the equation of motion of the particle. 1

(ii) Find the time taken by the particle to reach its highest point, and find the height of that point. 4

(iii) The particle falls to its original position under gravity and under the same law of resistance. Will the time of descent be greater or less than the time of ascent? Give reasons for your answer. 3

(b) A region R is bounded by the curves $y = 12 - x^3$ and $y = 12 - 4x$.

(i) Find the co-ordinates of the points of intersection of these curves and hence draw a neat sketch showing the resultant regions. 4

(ii) By taking cylindrical shells of width Δx and radius x , show that the volume generated by revolving R about the y -axis is $\frac{256\pi}{15}$ units³. 3

QUESTION 3 (START A NEW PAGE) (15 Marks)**Marks**

(a) The section of a solid cut by any plane perpendicular to the x -axis is a square with the ends of a diagonal lying on the parabolas $y^2 = 9x$ and $x^2 = 9y$.

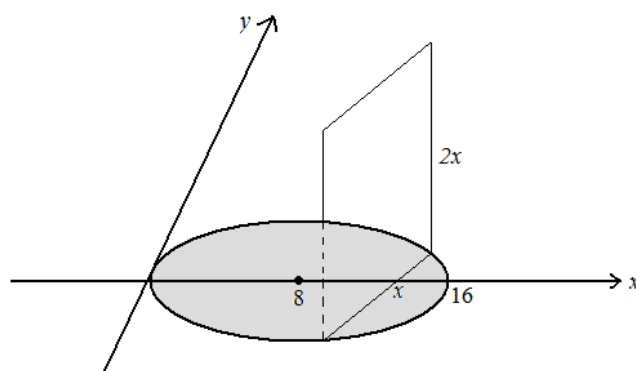
(i) Find the points of intersection of the two parabolas. 2

(ii) Show that the area $A(x)$ is given by:

$$A(x) = \frac{1}{162} \left[(27)^2 x - 54x^{\frac{5}{2}} + x^4 \right] \quad 4$$

(iii) Find the volume of the solid. 4

(b) The base of a solid is the circle $x^2 + y^2 = 16x$, and every planar section perpendicular to the x -axis is a rectangle whose height is twice the distance of the plane of the section from the origin as shown in the diagram:



(i) Show that the area $A(x)$ is given by: 2

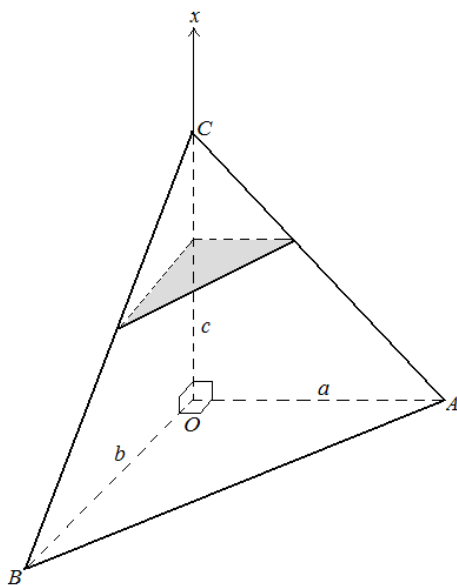
$$A(x) = 4x\sqrt{64 - (x - 8)^2}$$

(ii) Find the volume of the solid. 3

QUESTION 4 (START A NEW PAGE) (15 Marks)**Marks**

- (a) The tangent at $P(\tan \theta, \sec \theta)$ on the hyperbola $y^2 - x^2 = 1$ cuts the x and y axes at X and Y respectively.
- (i) Show this hyperbola is rectangular. **1**
- (ii) Find the coordinates of X and Y . **2**
- (iii) Through X and Y lines are drawn parallel to the co-ordinate axes to intersect at Q . Find the coordinates of Q . **1**
- (iv) Hence, find the cartesian equation of the locus Q as P varies. **2**
- (v) Sketch the graph of the locus. **2**

- (b) The tetrahedron formed by three mutually perpendicular edges of lengths a, b, c is shown in the diagram below:



Let the origin be the intersection of the edges, and let the x -axis lie along the edge of the length c as shown in the diagram. A typical cross section is a right triangle with legs of length d and e , parallel respectively to the edges of lengths a and b .

- (i) Using similar triangles, show that the area of the cross-section $A(x)$ is given by:

$$A(x) = \frac{ab}{2c^2} (c - x)^2$$

4

- (ii) Find the volume of the solid generated. **3**

Question (1)

(i) $xy = a^2, y = a^2/x$

$$\therefore \frac{dy}{dx} = -a^2/x^2$$

$$\left. \frac{dy}{dx} \right|_{x=at} = \frac{-a^2}{a^2t^2} = -\frac{1}{t^2}$$

$\therefore$ Equation of $tg t$

$$at + P$$
$$y - \frac{a}{t} = -\frac{1}{t^2}(x - at)$$

$$x + t^2y - 2at = 0$$
$$\left(\frac{1}{t^2}\right)x + y = \frac{2a}{t}$$
$$y = \left(-\frac{1}{t^2}\right)x + \frac{2a}{t}$$

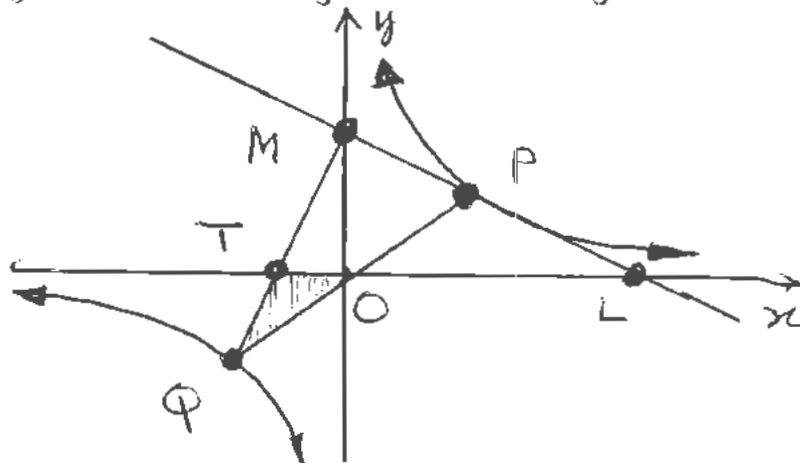
Gradient
of $tg t$
 $= -1/t^2$

[1]

Correct
equation

[1]

(ii) $\Phi(-at, -a/t)$ by
symmetry of hyperbola.



When $x = 0, y = \frac{2a}{t}$

$$\therefore M\left(0, \frac{2a}{t}\right)$$

Coordinates
of M
 $\left(0, \frac{2a}{t}\right)$

[1]

$$m_{MQ} = \frac{2a/t + a/t}{0 + at} = \frac{3a}{t^2}$$

$$= \frac{3}{t^2}$$

∴ Equation of MQ is:

$$y - \frac{2a}{t} = \frac{3}{t^2} (x - 0)$$

$$3x - t^2 y + 2at = 0$$

$$y = \left(\frac{3}{t^2}\right)x + \frac{2a}{t}$$

When $y = 0$, MQ
meets x-axis at T

$$\therefore -\frac{2a}{t} = \frac{3x}{t^2}$$

$$\Rightarrow x = -\frac{2at}{3}$$

$$\therefore T \left(-\frac{2at}{3}, 0\right)$$

i.e. $|OT| = \frac{2at}{3}$ unit

Area of ΔQOT

$$= \frac{1}{2} |OT| \cdot |y \text{ coord } Q|$$

$$= \frac{1}{2} \times \frac{2at}{3} \times \frac{a}{t}$$

$$= \frac{a^2}{3}$$

Correct
equation
of MQ

[1]

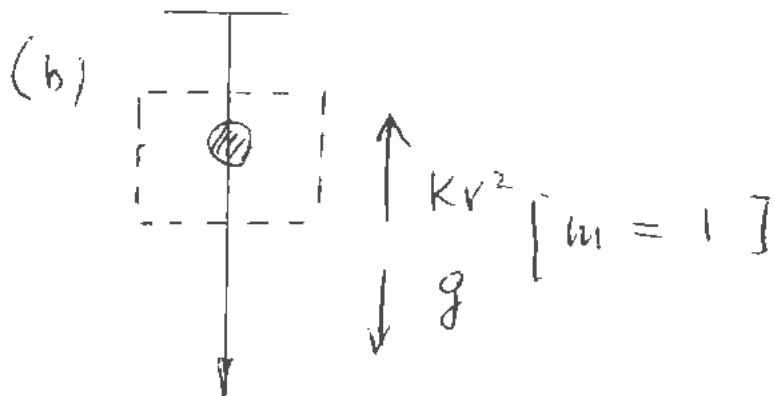
Coordinates
of T
 $\left(-\frac{2at}{3}, 0\right)$

[1]

$$\frac{a^2}{3} \quad [1]$$

Incorrect
expression
for $|OT|, |y|$

$\left[\frac{1}{2}\right]$



Taking positive to be 'down'

$$\ddot{x} = g - kv^2$$

$$v \frac{dv}{dx} = g - kv^2$$

$$\frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\int dx = \int \frac{v dv}{g - kv^2}, \quad \int dx = -\frac{1}{2k} \int \frac{-2kv dv}{g - kv^2}$$

$$x = -\frac{1}{2k} \ln |g - kv^2| + c$$

When $x = 0$, $v = 0$

$$\Rightarrow c = \frac{1}{2k} \ln g$$

$$\text{i.e. } x = \frac{1}{2k} \ln \left| \frac{g}{g - kv^2} \right|$$

When $x = s$

$$s = \frac{1}{2k} \ln \left| \frac{g}{g - kv^2} \right|$$

$$e^{2ks} = \frac{g}{g - kv^2}$$

$$g - kv^2 = g e^{-2ks}$$

$$v^2 = \frac{g}{k} (1 - e^{-2ks})$$

$$\text{i.e. } v = \sqrt{\frac{g}{k} (1 - e^{-2ks})} \quad (v > 0) \quad \textcircled{1}$$

Correct form
of D.E

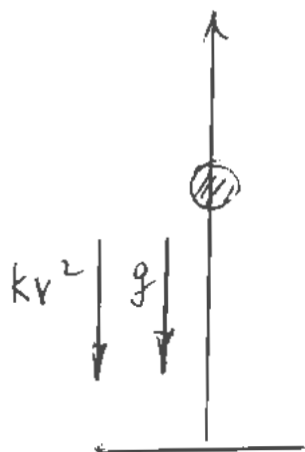
[1]

Correct
expression
for s
with approp.
initial condition.

[1]

Correct
expression
for v

[1]



Taking positive to be 'up'

$$\ddot{x} = -g - kv^2$$

$$v \frac{dv}{dx} = -(g + kv^2)$$

$$\begin{aligned} \int dx &= - \int \frac{v}{g + kv^2} dv \\ &= - \frac{1}{2k} \int \frac{2kv dv}{g + kv^2} \end{aligned}$$

$$\therefore x = - \frac{1}{2k} \ln(g + kv^2) + C$$

When $x = 0$, $v = u$

$$\Rightarrow C = \frac{1}{2k} \ln(g + ku^2)$$

$$\therefore x = \frac{1}{2k} \ln\left(\frac{g + ku^2}{g + kv^2}\right)$$

At max. height $v = 0$

$$x = \frac{1}{2k} \ln\left(\frac{g + ku^2}{g}\right)$$

Correct DE
and integral

$$\int dx = - \int \frac{v dv}{g + kv^2}$$

[1]

[1]

Correct
expression for x
before reaching
maximum height

[1]

Max height

$$v = 0$$

$$\therefore x = \frac{1}{2k} \ln\left(\frac{g + ku^2}{g}\right)$$

From equation (1) in (i)

$$v^2 = \frac{g}{k} (1 - e^{-2ks})$$

Rearranging,

$$e^{-2ks} = 1 - \frac{kv^2}{g} \quad \text{--- (3)}$$

From equation (2) in (ii)

$$s = \frac{1}{2k} \ln \left(\frac{g + kv^2}{g} \right)$$

Substitute s into (3)

and $v = w$, we have

$$e^{-\ln \left(\frac{g + kv^2}{g} \right)} = \frac{g - kw^2}{g}$$

$$\therefore \frac{g}{g + kv^2} = \frac{g - kw^2}{g}$$

$$g^2 = (g + kv^2)(g - kw^2)$$

$$gkv^2 = gkw^2 + k^2v^2w^2$$

$$v^2 = w^2 + \frac{k}{g}v^2w^2$$

$$v^2 = w^2 + \frac{v^2w^2}{v^2}$$

$$v^2v^2 = v^2w^2 + v^2w^2$$

$$w^2 = \frac{v^2v^2}{v^2 + v^2}$$

$$w^{-2} = \frac{v^2 + v^2}{v^2v^2}$$

$$\therefore w^{-2} = v^{-2} + v^{-2}$$

- Correct expression to start off
- or equivalent merit

[1]

Equivalent expression for e^{-2ks}

$$\text{i.e. } \frac{g}{g + kv^2} = \frac{g - kw^2}{g}$$

or equivalent merit

[1]

Algebraic manipulation to get to $w^{-2} = v^{-2} + v^{-2}$

[1]

2011 Y-12 T2

MATHEMATICS Extension 1: Question 2

Suggested Solutions

Marks

Marker's Comments

$$(i) \quad m\ddot{x} = -\frac{v}{c} - mg \quad \text{or} \quad m\ddot{x} = -m\frac{v}{c} - mg$$

$$\Rightarrow \ddot{x} = -\frac{v}{c} - g$$

$$(ii) \quad \int \frac{c dv}{v+cg} = -\int dt$$

$$c \ln(v+cg) = -t + k_1$$

$$* \quad t = c(\ln(h) - \ln(v+cg))$$

$$\text{max ht } v=0, \quad T = c \ln \frac{h}{g}$$

$$v \frac{dv}{dx} = -\left(\frac{cg+v}{c}\right)$$

$$\int_{c(h-g)}^0 \frac{v}{cg+v} dv = -\frac{1}{c} \int_0^h dx \quad (\text{max ht, } v=0)$$

$$-S = \int_{c(h-g)}^0 c - \frac{c^2 g dv}{v+cg}$$

$$S = c^2(h-g) + \left[c^2 g \ln(v+cg) \right]_{c(h-g)}^0$$

$$S = c^2(h-g) + c^2 g \ln \frac{g}{h}$$

$$\text{max or} \quad \text{From } * \quad v = e^{-\frac{t}{c}} \cdot ch - cg$$

$$\frac{ds}{dt} = e^{-\frac{t}{c}} \cdot ch - cg$$

$$s = -ch(e^{-\frac{t}{c}}) - cg + k_2$$

forgot max ht $v=0$
-1/2

2011 Yr 12 T2 MATHEMATICS Extension 2: Question 2

Suggested Solutions	Marks	Marker's Comments
$t=0, s=0, k_2=c^2h$ $S = -c^2h e^{-\frac{v}{c}} - cgt + c^2h$ at max ht $T = c \ln \frac{h}{g}$ $\text{max } S = c^2(h-g) + c^2g \ln(\frac{g}{h})$ iii) Time to reach max ht $= c \ln \frac{h}{g}$ Since $v = c(h-g) > 0 \therefore h > g$ Try to find the downward velocity at $t = c \ln \frac{h}{g}$ $\ddot{x} = g - \frac{v}{c}$ $\int_0^v \frac{dv}{g - \frac{v}{c}} = \int_0^{c \ln \frac{h}{g}} dt$ $\left[-c \ln(cg - v) \right]_0^v = c \ln \frac{h}{g}$ $\ln\left(\frac{cg}{cg-v}\right) = \frac{h}{g} \therefore \frac{cg}{cg-v} = \frac{h}{g}$ $v = \frac{cg(h-g)}{h} = cg\left(1 - \frac{g}{h}\right)$ Try to find distance travelled from $t=0$ to $t = c \ln \frac{h}{g}$ $v \frac{dv}{dx} = \frac{cg-v}{c} \therefore \int_0^{cg(1-\frac{g}{h})} \frac{v dv}{cg-v} = \int_0^x \frac{dx}{c}$ $cg\left(1 - \frac{g}{h}\right) \int_0^x \left(-1 + \frac{cg}{cg-v}\right) dv = \frac{x}{c}$	1 1 1m	Common mistakes. $-\int \frac{cv}{cg-v} dv = \int_0^s dx$ max 2m. $\ddot{x} = g - \frac{v}{c}$ most students get this $\frac{1}{2}m$ only when they do not successfully integrate any of the x, v, or t.

Prd

Ex 2 7/12 2011 Tr Q2 cat'd

P3

$$-cg(1 - \frac{g}{h}) - \left[cg \ln(cg - v) \right]_0^{cg(1 - \frac{g}{h})} = \frac{x}{c}$$

$$-cg(1 - \frac{g}{h}) - cg \ln\left(\frac{cg - cg + cg \frac{g}{h}}{cg}\right) = \frac{x}{c}$$

$$cg(\frac{g}{h} - 1) - cg \ln\left(\frac{g}{h}\right) = \frac{x}{c}$$

$$x = c^2 g \left(\frac{g}{h} - 1 \right) - c^2 g \ln\left(\frac{g}{h}\right) \quad |m$$

$$h) = \text{MAX } ht - x = c^2(h - g) - c^2 g \ln \frac{h}{g} - c^2 \left(\frac{g}{h} - g \right) + c^2 g \ln \frac{g}{h}$$

$$f(h) = c^2 h - \frac{c^2 g^2}{h} - 2c^2 g \ln \frac{h}{g}$$

$$f'(h) = c^2 + \frac{c^2 g^2}{h^2} - 2c^2 g \left(\frac{g}{h}, \frac{1}{g} \right) = c^2 \left(1 - \frac{2g}{h} + \frac{g^2}{h^2} \right) = c^2 \left(\frac{g}{h} - 1 \right)^2 > 0$$

Since $h > g$, $f'(x) > 0 \therefore \text{max } ht > x$

At $t = c \ln \frac{h}{g}$, has not reached pt of projection.

must have conclusion

$\therefore$ Time descent is greater than ascend

$\frac{1}{2}m$

many students try to find the time to reach the pt of projection

$$\frac{dv}{dt} = \frac{cg - v}{c}$$

$$\int_0^v \frac{dv}{cg - v} = \int_0^t \frac{dt}{c} \quad \ln\left(\frac{cg}{cg - v}\right) = \frac{t}{c} \quad \therefore t = c \ln\left(\frac{cg}{cg - v}\right) \quad |m$$

or try to find the velocity reaching pt of projection.

$$\frac{t}{c} = \ln \frac{cg}{cg - v} \quad v = cg(1 - e^{-t/c}) \quad |m$$

to relate to the distance $\frac{v dv}{dx} = \frac{cg - v}{c}$

$$\int_0^v \frac{v dv}{cg - v} = \int_0^s \frac{dx}{c} \quad \int \left(-1 + \frac{cg}{cg - v} \right) dv = \frac{s}{c}$$

$$s = - \left[cv + c^2 g \ln \frac{cg - v}{cg} \right] \quad |m$$

MATHEMATICS Extension 2: Question 2

Suggested Solutions

Marks

Marker's Comments

Q2(b)(i) $y = 12 - x^3$ — (1)
 $y = 12 - 4x$

$$12 - 4x = 12 - x^3$$

$$x^3 - 4x = 0$$

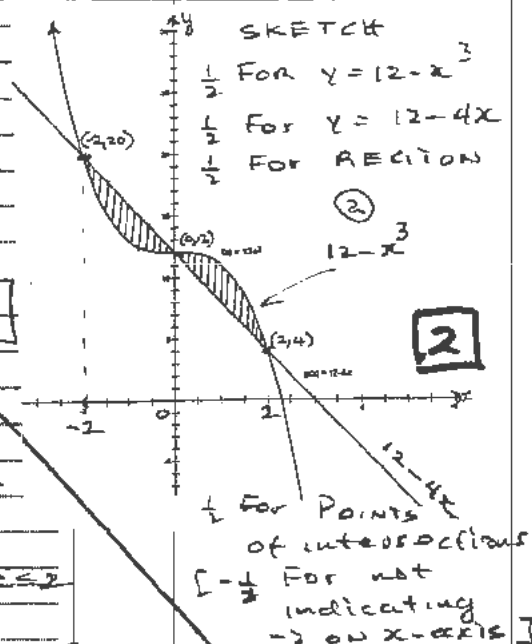
$$x(x^2 - 4) = 0$$

$$\therefore x = 0 \text{ or } \pm 2$$

Curve intersect

at $(-2, 20)$, $(0, 12)$ and $(2, 4)$ ✓

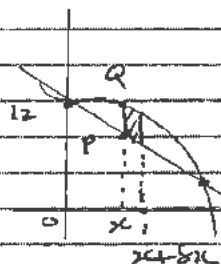
[2]



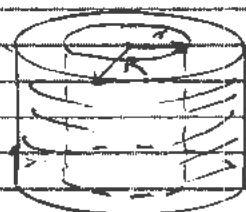
[2]

(ii) x the area bounded for $-2 \leq x \leq 0$ rotated about y-axis is identical to the area bounded for $0 \leq x \leq 2$, rotated about the y-axis

$$\therefore \text{Volume} = 2 \times \text{Volume } 0 \leq x \leq 2$$



Taking a slice of thickness δx parallel to the y-axis at x is rotated about the y-axis forming a cylindrical shell



APPROACH I

$$h = PQ = y_u - y_l = 12 - x^3 - (12 - 4x) = 4x - x^3$$

$$r = x$$

$$R = x + \delta x$$

$$A(x) = \pi [R^2 - r^2] = \pi (R + r)(R - r)$$

$$= \pi (2x + \delta x)(\delta x)$$

$$= \pi (2x \delta x + (\delta x)^2)$$

$$\therefore A(x) \approx 2\pi x \delta x, \text{ neglect } (\delta x)^2 \text{ as } (\delta x)^2 \neq 0$$

$$\text{Volume of shell } \delta V \approx 2\pi x \delta x \times h$$

$$\delta V \approx 2\pi x (4x - x^3) \delta x$$

$$\text{Volume of solid } V \approx 2 \times \sum_{x=-2}^2 2\pi x (4x - x^3) \delta x$$

$$V = 4\pi \lim_{n \rightarrow \infty} \sum_{k=1}^n x (4x - x^3) \delta x$$

$$V = 4\pi \int_{-2}^2 (4x^2 - x^4) dx$$

$$= 4\pi \left[\frac{4}{3} x^3 - \frac{1}{5} x^5 \right]_{-2}^2$$

$$V = 4\pi \left[\frac{32}{3} - \frac{32}{5} \right] = \frac{256\pi}{15}$$

(as $V_{\text{left}} = V_{\text{right}}$)

[max if do
 $\pi \int_{-2}^2 x^4 - 4x^2 dx$
 $= -128 \frac{\pi}{5}$

[3]

$$\therefore \text{VOLUME is } \frac{256\pi}{15} \pi^3$$

$$\text{OR } V = V_1 + V_2 \rightsquigarrow 2\pi \int_{-2}^0 x(y_u - y_l) dx + 2\pi \int_0^2 x(y_u - y_l) dx$$

For $\frac{256\pi}{15}$

MATH EXT 2, ASSESSMENT TEST 3 TERM 2, 2011

MATHEMATICS Extension 2: Question 3		Marks	Marker's Comments
$x^2 = 9y$ (i) $y^2 = 9x$ (1) $x^2 = 9y$ — (1) $y^2 = 9x$ — (2) $\text{sub (1)} \Rightarrow y = \frac{x^2}{9}$ — (1a) $\therefore y^2 = \frac{x^4}{81} = 9x$ $\therefore x^4 = 81 \cdot 9x = 729x$ $\therefore x(x^3 - 729) = 0$ $\therefore x = 0 \text{ or } x = 9$ $y = 0$ $y = 9$ TOP VIEW $\therefore$ Points of intersection $(0,0)$ and $(9,9)$			$y^2 = 9x$ $y^4 = 81x^2 = 81 \cdot 9y$ etc $\frac{1}{2}$ For correct equation $\frac{1}{2}$ For $(0,0)$ 1 For $(9,9)$ [2]
(ii) Diagonal $PQ = y_2 - y_1 = 3\sqrt{x} - \frac{x^2}{9}$ $= \frac{1}{9} (27\sqrt{x} - x^2)$ Area of Square = s^2 or $\frac{1}{2} PQ \times PQ$ ($\frac{1}{2}$ product of diagonals) $\therefore A(x) = \frac{1}{2} \left[\frac{1}{9} (27\sqrt{x} - x^2) \right]^2$ $= \frac{1}{2} \times \frac{1}{81} (27^2 x - 54x^{\frac{5}{2}} + x^4)$ $= \frac{1}{162} (27^2 x - 54x^{\frac{5}{2}} + x^4)$			$-\frac{1}{2}$ For $\frac{x^2}{9} - 3\sqrt{x} < 0$ For organising/starting For getting to $A(x)$ correctly [3]
(iii) - Volume of slice $\delta V \div A(x) \delta x$ - Volume of solid $V \div \sum \delta V (= \sum A(x_i) \delta x)$ $\therefore V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i) \delta x$ $\delta x \rightarrow 0$ ie $V = \int_0^9 \frac{1}{162} (27^2 x - 54x^{\frac{5}{2}} + x^4) dx$ $V = \frac{1}{162} \left[\frac{27^2 x^2}{2} - 54 \times \frac{2}{7} x^{\frac{7}{2}} + \frac{1}{5} x^5 \right]_0^9$ $= \frac{1}{162} \left[\frac{59049}{2} - \frac{236196}{7} + \frac{59049}{5} - 0 \right]$ $\left(= \frac{1}{162} [29524.5 - 33742.28... + 11809.8] \right)$ $V = \frac{6561}{140} = 46.864...$ $= 46 \frac{121}{140}$ $\therefore$ Volume of solid is $46 \frac{121}{140} u^3$			For organising/developing the set up to V For integrating correctly [4] $-\frac{1}{2}$ if $V = 46 \frac{121}{140} u^3$ Volume = $46 \frac{121}{140}$

(b)

A TOP VIEW

(i) $x^2 + y^2 = 16x$
 $x^2 - 16x + 64 + y^2 = 64$
 $(x-8)^2 + y^2 = 64$
 $y^2 = 64 - (x-8)^2$

upper semi-circle:
 $y = \sqrt{64 - (x-8)^2}$

$\therefore PQ = y_2 - y_1 = y - (-y) = 2y$

Area $A(x) = 2y \times 2x = 4xy$
 ie $A(x) = 4x\sqrt{64 - (x-8)^2}$

OR:
 Circle centre (8,0)
 radius 8 is
 $(x-8)^2 + y^2 = 64$

For getting/showing where
 $64 - (x-8)^2$
 comes from

1/2

1/2

2

(ii) volume of slice $\delta V \doteq A(x) \delta x$
 " of solid $V \doteq \sum_{i=1}^n A(x_i) \delta x$

ie $V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i) \delta x$
 $\delta x \rightarrow 0$

$V = \int_0^{16} 4x \sqrt{64 - (x-8)^2} dx$

METHOD I let $u = x-8$
 $x = u+8$
 $dx = du$

x	u
16	8
0	-8

$\therefore V = 4 \int_{-8}^8 (u+8) \sqrt{64-u^2} du$

$= 4 \left\{ \int_{-8}^8 u \sqrt{64-u^2} du + 8 \int_{-8}^8 \sqrt{64-u^2} du \right\}$

cos OPD FN $\left[-\frac{2}{3} (64-u^2)^{3/2} \right]_{-8}^8$

$= 4 \left\{ 0 + 8 \times \frac{1}{2} \text{area of circle} \right\}$

$= 4 \left\{ 0 + 8 \times \frac{1}{2} \times \pi \times 8^2 \right\}$

$= 4 \times 4 \times \pi \times 64$

$V = 1024\pi$

$\therefore$ volume of solid is 1024π

METHOD II $V = 4 \int_0^{16} x \sqrt{64 - (x-8)^2} dx = 4 \int_0^{16} x \sqrt{16x - x^2} dx$

$V = 4 \int_0^{16} (x-8+8) \sqrt{64 - (x-8)^2} dx = 4 \int_0^{16} -\frac{1}{2} (16-2x-16) \sqrt{16x-x^2} dx$

$= 4 \int_0^{16} (x-8) \sqrt{64 - (x-8)^2} dx + 32 \int_0^{16} \sqrt{16x-x^2} dx$

$= 4 \left[-\frac{1}{3} (64 - (x-8)^2)^{3/2} \right]_0^{16} + 32 \times \frac{1}{2} \times \pi \times 8^2$

$= 0 - 0 + 1024\pi = 1024\pi$

$\therefore V = 1024\pi$

METHOD IV IBPs

(1/2) For $\int_0^{16}$ if ..

METHOD II (harder way)

$x = 8 + 8 \sin \theta$
 $dx = 8 \cos \theta d\theta$

$V = 4 \cdot 8^3 \int_0^{\pi/2} (1 + \sin \theta) \cos^2 \theta d\theta$

Note: $\sqrt{\cos^2 \theta} = |\cos \theta| = \cos \theta$

$V = 2048 \int_0^{\pi/2} (\cos^2 \theta + \sin \theta \cos^2 \theta) d\theta$

$V = 2048 \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} + \sin \theta \cos^2 \theta \right) d\theta$

$= 2048 \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta - \frac{1}{3} \cos^3 \theta \right]_0^{\pi/2}$

$= 2048 \left[\frac{\pi}{4} + 0 - 0 - \left(-\frac{\pi}{4} + 0 - 0 \right) \right]$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

$V = 2048 \times \frac{\pi}{2} = 1024\pi$

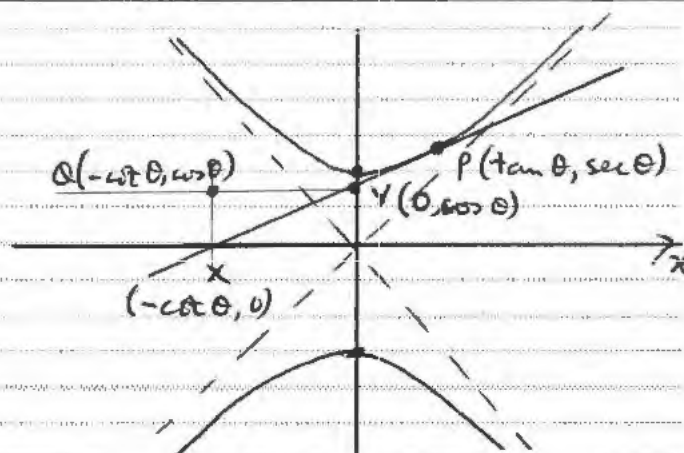
$V = 2048 \times \frac{\pi}{2} = 1024\pi$

Suggested Solutions

Marks

Marker's Comments

a)



i) Asymptotes given by

$$y^2 - x^2 = 0$$

$$y = \pm x$$

gradients are ± 1

Asymptotes are perpendicular so the given hyperbola is rectangular.

$$\text{ii) } 2y \frac{dy}{dx} - 2x = 0 \quad \frac{dy}{dx} = \frac{x}{y} = \frac{\tan \theta}{\sec \theta} = \sin \theta \text{ at } P$$

$$\therefore \text{Tangent at } P \text{ is } y - \sec \theta = \sin \theta (x - \tan \theta)$$

$$y = x \sin \theta + \cos \theta$$

$$\text{When } y = 0, x = -\cot \theta \quad \underline{X = (-\cot \theta, 0)}$$

$$x = 0, y = \cos \theta \quad \underline{Y = (0, \cos \theta)}$$

iii) From diagram $Q = (-\cot \theta, \cos \theta)$

$$\text{iv) } \left. \begin{array}{l} x = -\cot \theta \\ y = \cos \theta \end{array} \right\} \text{ since } \sec^2 \theta = 1 + \tan^2 \theta$$

$$\frac{1}{y^2} = 1 + \frac{1}{x^2}$$

Restrictions

$$\theta \neq \frac{\pi}{2}, \frac{3\pi}{2} \text{ (A at } \infty)$$

$$\theta \neq 0, \pi, \dots \text{ (Vertices)}$$

$$\underline{y^2 = \frac{x^2}{x^2 + 1}}$$

$$\Rightarrow x \neq 0, y \neq 0$$

One mark was given if the eccentricity was shown to be $\sqrt{2}$. (tho' this is a consequence rather than the definition)

Reasonable simplifⁿ was expected.

Distressing to see how many left $\frac{1}{\sec \theta}$ in working

Few gave the restrictions (none gave them in terms of θ). $\frac{1}{2}$ off for no restrictions.

Suggested Solutions

Marks

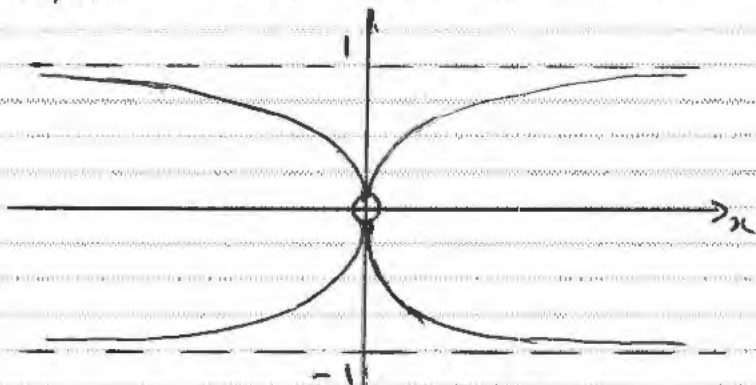
Marker's Comments

v) Symmetric in xy As $x \rightarrow \infty$, $y \rightarrow \pm 1$ (Horizontal asymptote)

$$2y \frac{dy}{dx} = -\frac{2}{x^3} \quad \therefore \frac{dy}{dx} = -\frac{1}{x^3 y}$$

 $\therefore$ As $(x, y) \rightarrow (0, 0)$, gradient increases without bound (i.e. vertical)

(0,0) is NOT allowed.

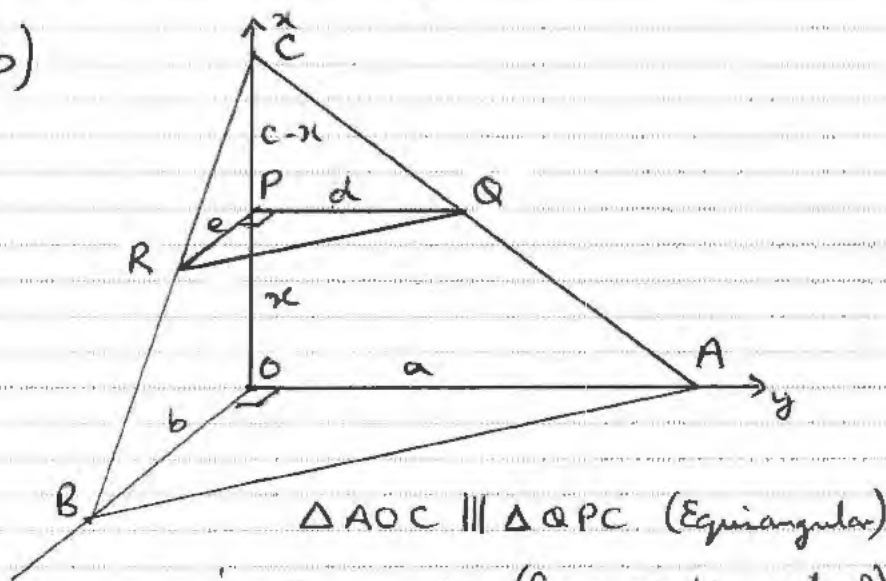


2

I wanted to mark
 $\frac{1}{2}$ horizontal asymp
 $\frac{1}{2}$ 4 way symmetry
 $\frac{1}{2}$ hole at (0,0)
 $\frac{1}{2}$ vertical asymp (0,0)

but I got a
 bit lenient with
 last 2 points.

b)

 $\triangle AOC \parallel \triangle QPC$ (Equiangular)

$$\therefore \frac{c}{c-x} = \frac{a}{d} \quad \left(\begin{array}{l} \text{Corresponding sides of} \\ \text{similar triangles in} \\ \text{the same ratio} \end{array} \right)$$

$$\therefore d = \frac{a(c-x)}{c}$$

Similarly, using triangles OBC and CPR,

$$e = \frac{b(c-x)}{c} \quad \therefore A(x) = \frac{ed}{2} = \frac{ab(c-x)^2}{2c^2}$$

4

Few people drew
 a decent diagram

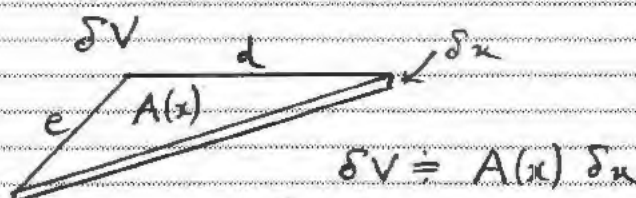
$\frac{1}{2}$ marks were
 given for similar
 triangle and its
 trappings (No
 full proof reqd.)

Diagrams
 generally
 very poor and
 scrappy (maybe
 time issue)

Suggested Solutions

Marks

Marker's Comments



$$\delta V = A(x) \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^c A(x) \delta x$$

$$= \lim_{\delta x \rightarrow 0} \sum_{x=0}^c \frac{ab(c-x)^2 \delta x}{2c^2}$$

$$= \int_0^c \frac{ab(c-x)^2}{2c^2} dx$$

$$= \frac{ab}{2c^2} \int_0^c (c-x)^2 dx$$

$$= \frac{ab}{2c^2} \left[-\frac{(c-x)^3}{3} \right]_0^c$$

$$= \frac{ab}{2c^2} \left(0 - \left(-\frac{c^3}{3} \right) \right)$$

$$= \frac{abc^3}{6c^2} = \frac{abc}{6}$$

∴ Volume is $\frac{abc}{6}$ cubic units

1/2

More diagrams would have been nice.

1/2

1/2 mark each for introductory infinitesimals.

1

1